Simulating Lattice Gauge Theories with Neural Networks

PHYS170 Machine Learning

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Motivation

- Confinement in QCD: fermions always exist in bound states (*mesons*) – no free quarks!

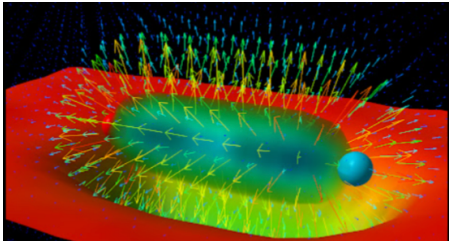


Figure: A bound state of a quark/anti-quark (fermions) pair held together by a gluon (boson)

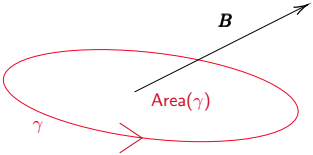
- QCD is hard: easier to analyze fermion-free theory
- "Pure gauge theory" $\Rightarrow$ bosonic loops instead of quark-antiquark pair bound by bosons!

Simplest example: 2D Electromagnetism

- $\text{Pure U}(1)\text{ Gauge Theory in 2D} \iff$ QM + 2D Maxwell theory of E&M
- Consider a loop (aka "string") γ enclosing area $\text{Area}(\gamma)$:

$$\mathcal{W}(\gamma) = \exp\left(i \int_{\text{Area}(\gamma)} \mathbf{B} \cdot d\,\mathbf{S}\right) \qquad \text{"Wilson loop" (Operator as in QM)}$$

- What is confinement in pure gauge theory?



$$\text{Answer: Theory is said to } \textbf{confine} \text{ if } \langle 0|\mathcal{W}(\gamma)|0\rangle \sim \mathrm{e}^{-\sigma \text{Area}(\gamma)} \text{ for some } \sigma > 0$$

σ = tension of the "string" γ, and $|0\rangle$ is the ground state.

- Key step: calculate the ground state $|0\rangle$ (Goal of this talk)

Lattice Hamiltonian

- Hamiltonian heavily studied in recent years – crucial for quantum simulations
- Hamiltonian proposed by Kogut and Susskind on the lattice ($\hbar = c = 1$)

$$H = \underbrace{\frac{g^2}{2}\sum_r \boldsymbol{E}(r)^2}_{\text{Electric energy}} + \underbrace{\frac{1}{g^2}\sum_r [1 - \cos B(r)]}_{\text{Magnetic energy}}$$

  on an $L \times L$ spatial lattice with periodic boundary conditions
- Canonical commutation relations ($B = \partial_x \theta_y - \partial_y \theta_x$)

$$[E_x, \theta_x] = [E_y, \theta_y] = i \qquad \text{(analogous to } [x, p] = i\hbar \text{ with } \hbar = 1\text{)}$$

- Gauge invariance implies any eigenstate $H|\psi\rangle = E|\psi\rangle$ satisfies:

$$\nabla \cdot \boldsymbol{E}|\psi\rangle = 0 \qquad \text{(Gauss' law)}$$

Architecture Overview

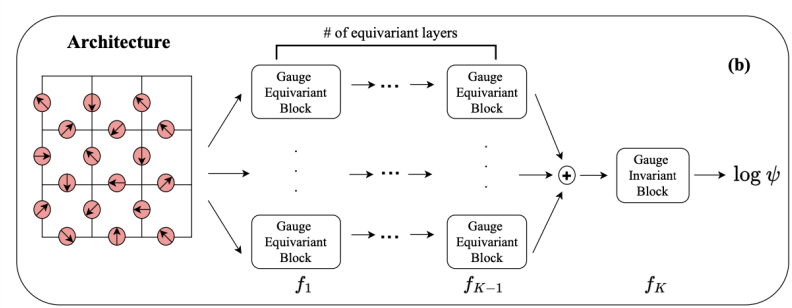


Figure: Model Architecture (Luo et al. 2022)

- **Design:** Neural network that respects Gauge invariance
- **Operation:** Gauge field configuration (θ) → Equivariant blocks → Invariant block → Log wave function amplitude $\psi(\theta)$

Feature Extraction

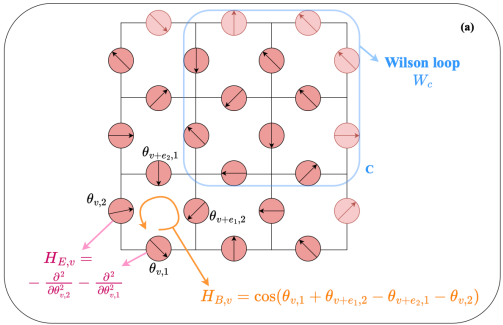


Figure: Lattice Configuration (Luo et al. 2022)

- **Lift Layer:** Maps Gauge links $U_{v,\mu} = e^{i\theta_{v,\mu}}$ to multiple channels $(\tilde{U}_{v,\mu,f},\ f \in (1, ..., F))$
- **Importance:** $(\tilde{U}_{v,\mu,f})$ are higher-dimensional yet transform as Gauge links for each μ
- **Plaquettes:** Build Gauge invariant features by multiplying $\tilde{U}_{v,i,f}$ around lattice loops
- **Importance:** Features built from Plaquettes in later layers are also Gauge Invariant

Gauge Equivariant Block

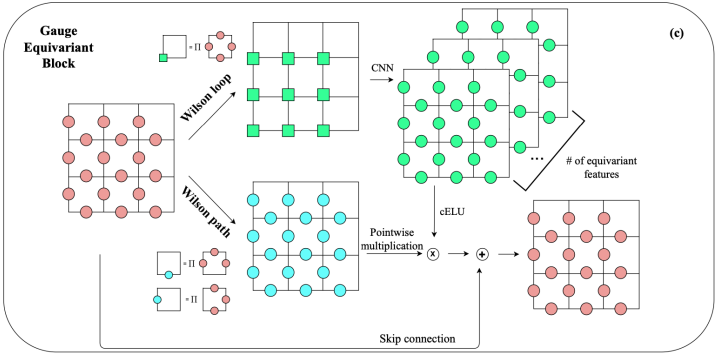


Figure: Gauge Equivariant Layers (Luo et al. 2022)

- **CNN Layers:** Extracts spatial correlations on plaquettes as modulation signals M
- **Residual Connection:** $\tilde{U} \leftarrow \tilde{U}(1 + cELU(M))$, cELU(x)$\neq 0$ for $x < 0$: better grad. descent
- **Importance:** M is Gauge Invariant so Gauge properties of $\tilde{U}$ are preserved

Gauge Invariant Block (Figure 1d)

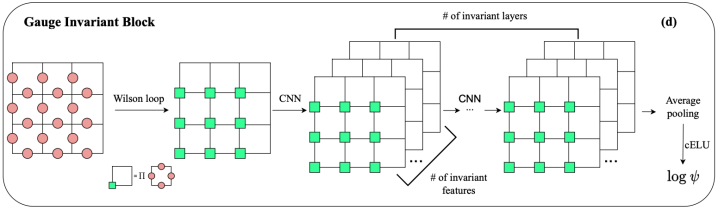


Figure: Gauge Invariant Layer (Luo et al. 2022)

- **Complex CNN:** Computes Gauge invariant plaquettes (P_{final}) from final link ($\tilde{U}$) values
- **Spatial Pooling:** Averages (P) over lattice sites and produces summary statistics ($\bar{z}$)
- **Wavefunction Calculation:** Uses linear layer to predict wavefunction evaluated for link configuration (θ) from $\bar{z}$
- **Importance:** Ensures wave function is Gauge and translation invariant

Variational Monte Carlo

Variation principle

To find the ground state: given ψ_α from the model, we find the model parameters α (weight & biases) that minimises

$$\overline{\alpha} = \arg\min_\alpha \frac{\langle \psi_\alpha | \hat{H} | \psi_\alpha \rangle}{\langle \psi_\alpha | \psi_\alpha \rangle} \tag{1}$$

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where $E^\alpha_{loc}(\theta) = \frac{\hat{H}\psi_\alpha(\theta)}{\psi_\alpha(\theta)}, \qquad \langle\langle \cdot \rangle\rangle = \int \cdot \, |\psi(\theta)|^2 \mathrm{d}\theta$

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Optimization

$$\text{loss} = \left\langle\!\!\left\langle \left[\tilde{E}_{loc}(\theta) - \langle\!\langle \tilde{E}_{loc}(\theta) \rangle\!\rangle \right] \log \psi_\alpha(\theta) \right\rangle\!\!\right\rangle \tag{5}$$

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Optimization loop

Draw ψ_α from model → Draw θ using ψ_α → Using θ, ψ_α compute loss

→ Gradient-descent: update α → Draw ψ_α from model → ⋯

Results

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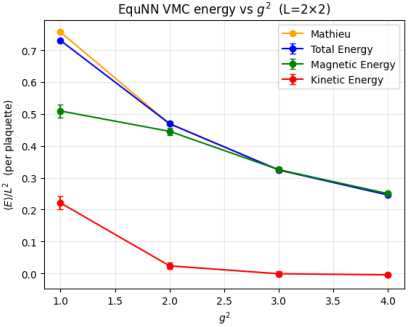
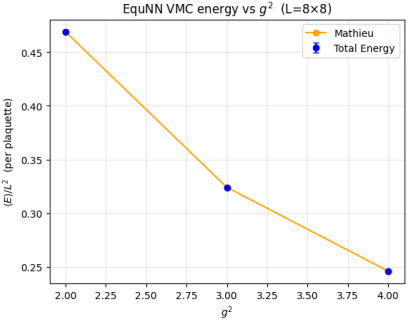
- This solution uses the variational principle as well, but does not use any machine learning.
- Finally, we have the loss function, which, when minimized will give the ground state.

Results

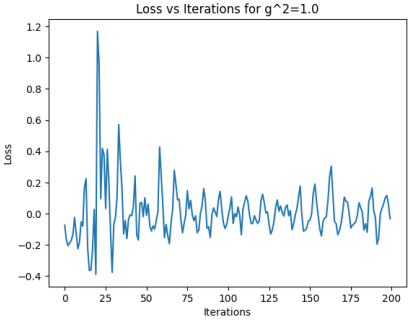


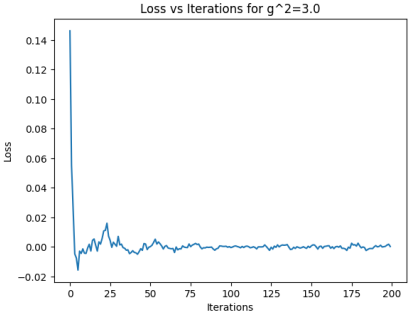


- Our energies have the same functional form as the analytical 1x1 solution as a function of the coupling (good)!
- Our energies match better in the strong coupling limit.

Results (continued)

- Training converges very rapidly for large values of g, slower convergence for small g.





References I

- Bender, Julian et al. (Oct. 2020). "Real-time dynamics in compact QED using complex periodic Gaussian states". In: *Physical Review Research* 2.4.
- Luo, Di et al. (2022). "Gauge Equivariant Neural Networks for 2+1D U(1) Gauge Theory Simulations in Hamiltonian Formulation". In: *arXiv:2211.03198*.

Acknowledgements

Thanks to Prof. Do, Gabi and Morgan for the class and helping us along the project!

Thank you!

Supplementary slides

Variational Principle Details

This can be recast in terms of the quantities accessible in the numerical sampling of angle configurations.

$$\nabla_\alpha \left[\frac{\langle \psi_\alpha | \hat{H} | \psi_\alpha \rangle}{\langle \psi_\alpha | \psi_\alpha \rangle} \right] = \frac{\langle \nabla_\alpha \psi_\alpha | \hat{H} | \psi_\alpha \rangle}{\langle \psi_\alpha | \psi_\alpha \rangle} - \frac{\langle \psi_\alpha | \hat{H} | \psi_\alpha \rangle}{\langle \psi_\alpha | \psi_\alpha \rangle} \frac{\langle \nabla_\alpha \psi_\alpha | \psi_\alpha \rangle}{\langle \psi_\alpha | \psi_\alpha \rangle} + h.c. \tag{6}$$

Using

$$\nabla_\alpha \psi_\alpha(\theta) = \psi_\alpha(\theta) \nabla_\alpha \log \psi_\alpha(\theta) \tag{7}$$

the first term can be written as

$$\begin{aligned} \frac{\langle \nabla_\alpha \psi_\alpha | \hat{H} | \psi_\alpha \rangle}{\langle \psi_\alpha | \psi_\alpha \rangle} &= \frac{\int \mathrm{d}\theta \ \nabla_\alpha \psi_\alpha^*(\theta) \hat{H} \psi_\alpha(\theta)}{\int \mathrm{d}\theta \ |\psi_\alpha(\theta)|^2} \\ &= \frac{\int \mathrm{d}\theta \ \left(\nabla_\alpha \log \psi_\alpha^*(\theta) \right) \psi_\alpha^*(\theta) E_{loc}(\theta) \psi_\alpha(\theta)}{\int \mathrm{d}\theta \ |\psi_\alpha(\theta)|^2} \\ &= \langle\!\langle \nabla_\alpha \log \psi_\alpha^*(\theta) E_{loc}(\theta) \rangle\!\rangle \end{aligned} \tag{8}$$

The second part of the second term can be recast as

$$\frac{\langle \nabla_\alpha \psi_\alpha | \psi_\alpha \rangle}{\langle \psi_\alpha | \psi_\alpha \rangle} = \frac{\int \mathrm{d}\theta \ \nabla_\alpha \log \psi_\alpha^*(\theta) \ |\psi_\alpha(\theta)|^2}{\int \mathrm{d}\theta \ |\psi_\alpha(\theta)|^2} = \langle\!\langle \nabla_\alpha \log \psi_\alpha^*(\theta) \rangle\!\rangle \tag{9}$$

The first part of the second term is

$$\begin{aligned} \frac{\langle \psi_\alpha | \hat{H} | \psi_\alpha \rangle}{\langle \psi_\alpha | \psi_\alpha \rangle} &= \frac{\int \mathrm{d}\theta \ \psi_\alpha^*(\theta) \hat{H} \psi_\alpha(\theta)}{\int \mathrm{d}\theta \ |\psi_\alpha(\theta)|^2} \\ &= \frac{\int \mathrm{d}\theta \ E_{loc}(\theta) |\psi_\alpha(\theta)|^2}{\int \mathrm{d}\theta \ |\psi_\alpha(\theta)|^2} \\ &= \langle\!\langle E_{loc}(\theta) \rangle\!\rangle \end{aligned} \tag{10}$$

Plugging Eqs. (8)-(10) into Eq. (6), the gradient is

$$\nabla_\alpha \left[\frac{\langle \psi_\alpha | \hat{H} | \psi_\alpha \rangle}{\langle \psi_\alpha | \psi_\alpha \rangle} \right] = \langle\!\langle E_{loc}^\alpha(\theta) \nabla_\alpha \log \psi_\alpha^*(\theta) \rangle\!\rangle - \langle\!\langle E_{loc}^\alpha(\theta) \rangle\!\rangle \langle\!\langle \nabla_\alpha \log \psi_\alpha^*(\theta) \rangle\!\rangle + h.c. \tag{11}$$